

PART 3

Infrastructure & Investment Cost Estimation

Estimating the cost of extending Google Search and Sarvam AI to a new language, benchmarked against the Bengali corpus analysed in Parts 1–2

1. Objective

The goal of this section is to estimate the infrastructure and investment cost required for two contrasting classes of language technology — a generative large language model (Sarvam AI) and a web-scale retrieval system (Google Search) — to add support for a new language. The estimate is anchored to the same Bengali corpus used for the Zipf's Law and Heaps' Law analysis in Parts 1 and 2, so that the cost of processing that corpus can be compared directly against the scale of investment each production system would require.

2. Costing Methodology & Assumptions

All estimates are built on a single extrapolated computational baseline:

$$\text{Cost} = (\text{Word Count} / 100,000) \times \$1,000$$

This baseline converts raw corpus/training-data size into a processing cost and is applied consistently across all three scenarios. On top of this data-processing cost, each production scenario adds fixed cost components that reflect the parts of the pipeline that do not scale linearly with word count — compute/training engineering and evaluation for a generative model, and crawling/serving infrastructure for a retrieval system. These fixed components are treated as flat allocations rather than being derived from the corpus itself, since they represent one-time or infrastructure-level investments rather than per-word processing cost.

- **Current corpus baseline:** The current Bengali corpus (17,586,583 words) is used as the reference point for actual processing cost.
- **Sarvam AI data scale:** A training corpus of 4 billion words is assumed for the Sarvam AI generative LLM scenario, reflecting the scale of data a foundation model typically needs to reach usable quality in a new language.
- **Google Search data scale:** A corpus of 200 million words is assumed for the Google Search retrieval scenario, reflecting that indexing systems need broad web coverage rather than the deep, repeated exposure a generative model needs during training.
- **Scope:** Costs are theoretical and directional; they are meant to illustrate the relative order of magnitude and cost structure of the two architectures, not to serve as a vendor-accurate quote.

3. Summary of Results

Scenario	Estimated Cost (USD)
Current CSV file processing (Bengali corpus, 17,586,583 words)	\$175,865.83

Scenario	Estimated Cost (USD)
Sarvam AI — Generative LLM (new-language support)	\$40,594,000.00
Google Search — Retrieval (new-language support)	\$20,000,000.00

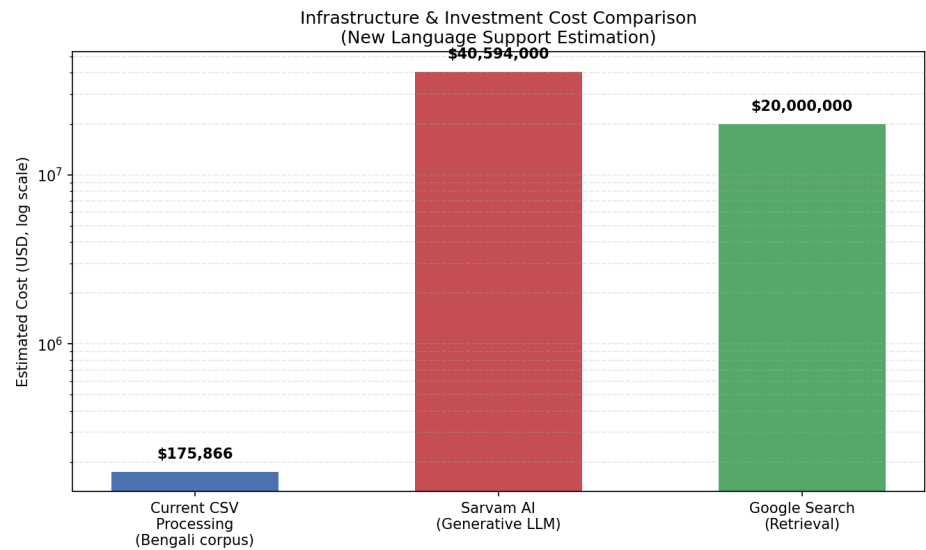


Figure 1: Estimated cost by scenario (log scale) — current corpus processing vs. full new-language investment for Sarvam AI and Google Search.

4. Cost Breakdown by Component

Breaking each total down into its components shows where the money actually goes for each type of system, and why the two totals differ by a factor of ~2x despite Google's raw data requirement being 20x smaller than Sarvam's.

4.1 Sarvam AI (Generative LLM)

Cost Component	Basis	Amount (USD)
Training-data acquisition & preparation	4,000,000,000 words @ \$1,000 / 100,000 words	\$40,000,000.00
Model training / compute engineering	Fixed allocation	\$150,000.00
Evaluation & safety testing	Fixed allocation	\$144,000.00
Deployment & serving infrastructure	Fixed allocation	\$300,000.00
Sarvam AI — Total		\$40,594,000.00

Over 98% of the Sarvam AI estimate (\$40,000,000 of \$40,594,000) comes directly from the data cost of training a foundation model, assumed here at 4 billion words. A generative model does not simply store text — every token is processed repeatedly across GPU clusters during training to learn the statistical structure of the language, which is why the per-word cost baseline is applied against such a large assumed corpus. The remaining \$594,000 covers training/compute engineering, evaluation and safety testing, and deployment infrastructure — comparatively small once the model exists, since serving a trained model is far cheaper than training it.

4.2 Google Search (Retrieval)

Cost Component	Basis	Amount (USD)
Data / indexing pipeline	200,000,000 words @ \$1,000 / 100,000 words	\$2,000,000.00
Crawling & indexing infrastructure	Fixed allocation	\$3,000,000.00
Edge caching & global load balancing	Fixed allocation	\$15,000,000.00
Google Search — Total		\$20,000,000.00

Google's estimate inverts the cost structure: only \$2,000,000 of the \$20,000,000 total (10%) comes from the data/indexing cost, since a search engine indexes text rather than training on it repeatedly, and therefore needs far less raw data (200 million words assumed) to add coverage of a new language. The bulk of the cost — \$18,000,000 — is ongoing infrastructure: crawling and indexing compute to keep the index fresh, and edge caching plus global load balancing to deliver sub-second results worldwide, 24/7.

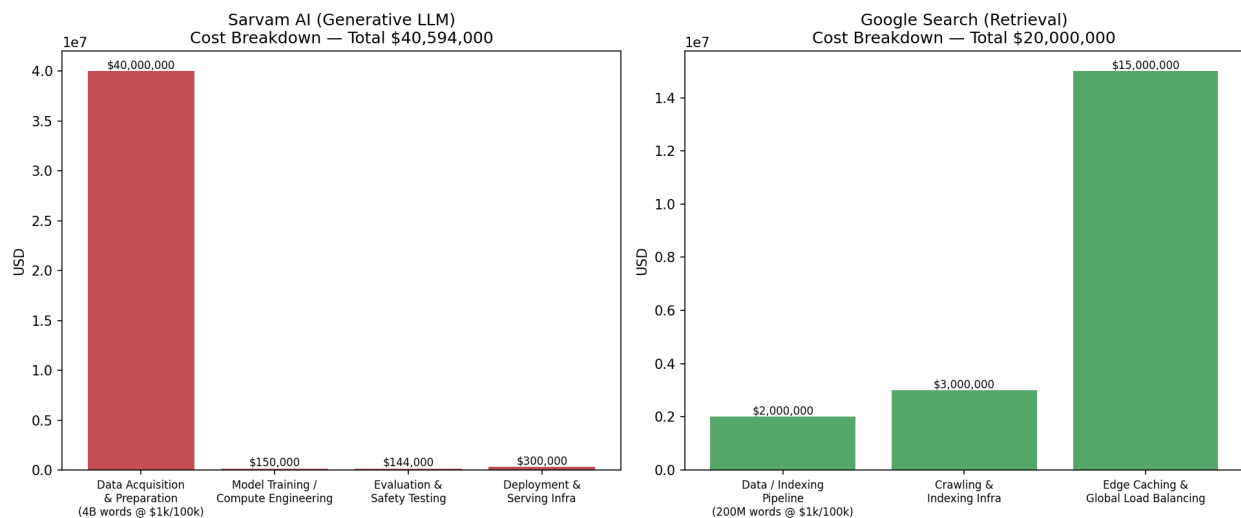


Figure 2: Component-level cost breakdown for Sarvam AI (left) and Google Search (right).

5. Interpretation

- **Where the cost lives** — Generative LLM (Sarvam AI): cost is front-loaded and dominated by data acquisition and training compute. Adding a new language means training on billions of words, so cost scales with training-corpus size far more than with the size of any single evaluation corpus.

- **Where the cost lives** — Retrieval (Google Search): cost is dominated by standing infrastructure — crawling, indexing, caching, and global serving — rather than by the data volume itself. Adding a new language mainly means extending crawl coverage and index support, which is comparatively cheap next to the infrastructure needed to serve it globally.
- **Scale gap** — At \$175,865.83, processing the current Bengali corpus is roughly 0.4% of the Sarvam estimate and 0.9% of the Google estimate — underscoring that the corpus-level analysis in Parts 1–2 sits at a very different scale from what a production system would need to actually support a new language end-to-end.
- **Why Sarvam isn't 20x more expensive** — Despite the Sarvam training corpus being 20x larger than Google's assumed retrieval corpus, Sarvam's total cost is only ~2x Google's — because Google's cost is driven by expensive, non-data-proportional infrastructure (crawling + global edge delivery), while Sarvam's cost is driven almost entirely by the linear data-cost term.

6. Conclusion

Under the \$1,000-per-100,000-words baseline, extending a generative LLM like Sarvam AI to a new language is estimated at roughly \$40.6 million, versus roughly \$20 million for a retrieval system like Google Search — about double the investment. The two architectures arrive at their totals through very different cost structures: Sarvam's cost is almost entirely a function of training-data volume, while Google's is dominated by the fixed, ongoing infrastructure needed to crawl, index, and serve results globally. This suggests that the generative-AI approach to new-language support is more data-hungry and front-loaded, whereas the retrieval approach is more infrastructure-heavy and ongoing — a distinction worth keeping in mind when comparing the two as strategies for expanding language coverage.